Srinivas Lakshmanan

Simulation Systems Architect



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Overview

Professional Summary:





ABOUT ME:

Simulation Systems Architect with 8+ years of experience architecting Multiphysics CAE simulation frameworks across rigid body dynamics, flexible bodies, contact mechanics, electromechanical systems, hydraulic actuation, controls simulation, and battery electrical/thermal modelling.
Worked with multiple CAE ISVs – Siemens, Hexagon MSC Software/Cadence, Gamma Technologies(GT-ISE), MATLAB/Simulink in roles spanning developer, tool designer, application engineer, pre-sales, post-sales, customer success, and product architect enabling both product innovation and customer adoption.

EDUCATION:

- Master of Science, Mechanical Engineering, University of Illinois at Chicago (2016-2018)
  Multibody Dynamics – Control Systems – Mechatronics – Robotics
- Bachelor of Engineering, Mechanical Engineering, Thiagarajar College of Engineering, India (2012-2016)
  Vehicle Dynamics, FEA, CFD

Work Experience:

- Product Architect, Caterpillar, Chicago (Nov 2023 – Current)
- Customer Success Engineer III – MSC Software/Hexagon Manufacturing Intelligence, Chicago (Oct 2022 – Oct 2023)
- Senior Product Application Engineer, Gama Technologies, Chicago (Oct 2019 – Oct 2022)
- Vehicle Dynamics Simulation Engineer, Chicago (July 2018 – Oct 2019)

CAPABILITIES

Core Competencies

Simulation Integration

Integrating simulation workflows to solve complex problems

CAE Application

Track record of applying CAE tools to solve real world problems

Architecting

Enterprise level simulation tool framework architecting using multiple CAE ISV tools.

Digital Twin development

Successfully implemented several digital twins in 8+ years

Programming Languages

Python, C++, NVIDIA openUSD certified, Omniverse, and MATLAB/Simulink

Exploring new frameworks

Quick learner capable of learning new simulation frameworks and methodologies with minimal guidance.

TECHNICAL STACK

Tools & Technologies

SIMULATION ENGINES

Omniverse, Isaac Sim, MSC Adams, Simcenter Motion, Siemens AMEsim, MSC Nastran, ANSYS Discovery, Project Chrono ANSYS Twin builder, MATLAB/Simulink, GT-ISE

MODELING TOOLS

NX CAD, NX Heeds, Pro Engineer, Solidworks

PROGRAMMING LANGUAGES

Python, C++, MATLAB, OpenUSD

PLATFORMS & OS

LINUX, ROS

Simulation APIs

MATLAB Python Engine, PyAnsys, AMEsim python API

VISUALIZATION

Omniverse, Simcenter, MSC Apex, Simulink

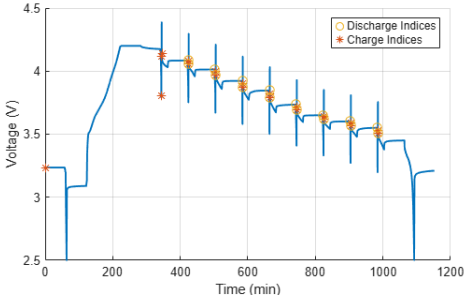
Projects

Notable Professional Projects:

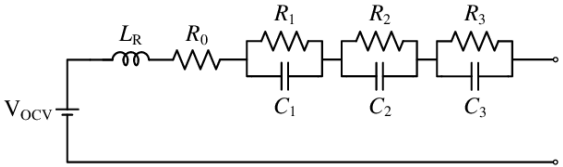
a) Battery Electrical and Thermal Model - Caterpillar Inc
b) Electric Powertrain Motor selection framework – Caterpillar Inc
c) Vehicle lateral and longitudinal motion control – Gamma Technologies LLC
d) Kinematics and Compliance – Suspension Model – Gamma Technologies LLC
e) Vehicle Steering System Controls Integration – MSC Software/Hexagon MI
f) Sprayer Dynamics and Hydraulics Simulation – CNH Industrial
g) Skid Steer Inverse Dynamics – CNH Industrial

a) Battery Electrical and Thermal Model

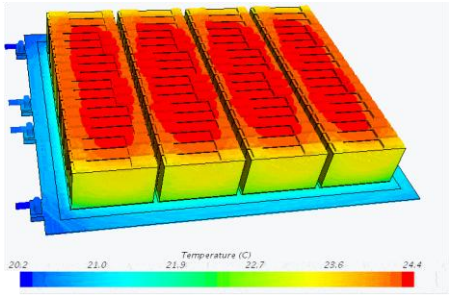
Caterpillar Inc



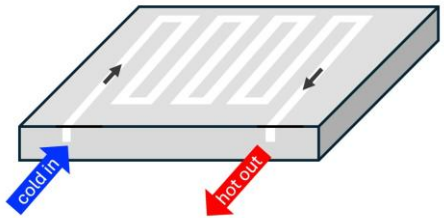
Use HPPC characterization data to identify battery 3RC model parameters



Build a battery Pack and cold plate thermal resistance network



Optimize thermal ROM using Ansys Discovery 3D CFD ambient heat rejection model, and machine test data



Built battery electrical and thermal models using a comprehensive workflow
Battery Types: LTO, LFP, NMC, cylindrical, prismatic, and blade cells.
Tools Used: Ansys Discovery, MATLAB/Simulink

b) Motor Selection Framework

Caterpillar Inc

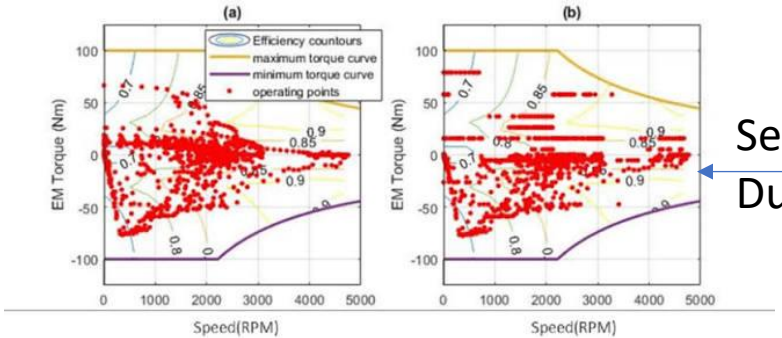


Size motor requirement based on machine requirements

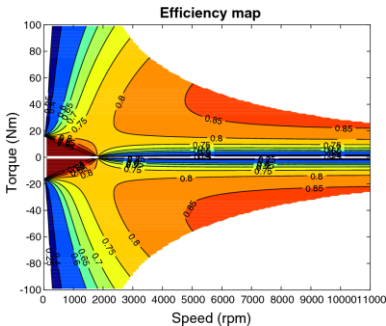


Identify feasible motors, and generate performance maps using Motor modelling tools – AMEsim, Motor CAD Simulink ROMS

Tools Used: Simcenter Studio, AMEsim, Simulink, MATLAB-Python Engine, MotorCAD python API.



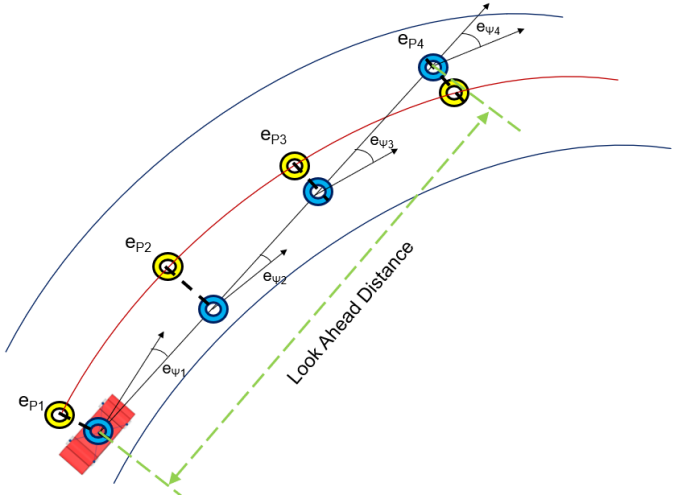
Select motor based on Duty cycle operating points



c) Vehicle Lateral and Longitudinal Motion Control

Gamma Technologies LLC

Lateral Motion Error Signals



Driver Pedal Dynamics (GG Diagram)

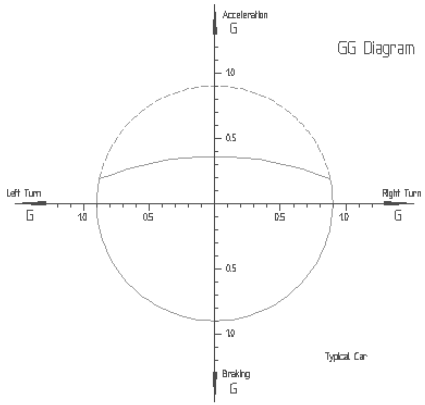


Figure 1

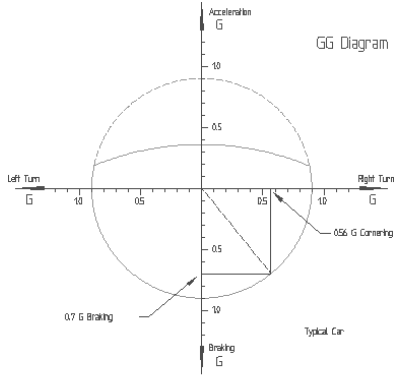


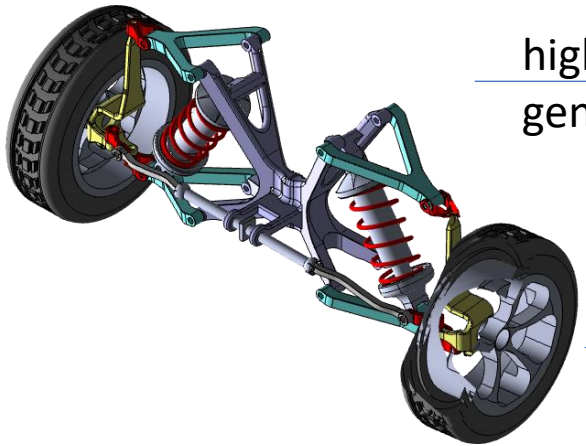
Figure 8

Developed vehicle lateral, and longitudinal motion controller that uses heading angle error, path deviation error , and tire friction circle diagrams as inputs.

Tools Used: GT-ISE

d) Kinematics and Compliance Suspension Model

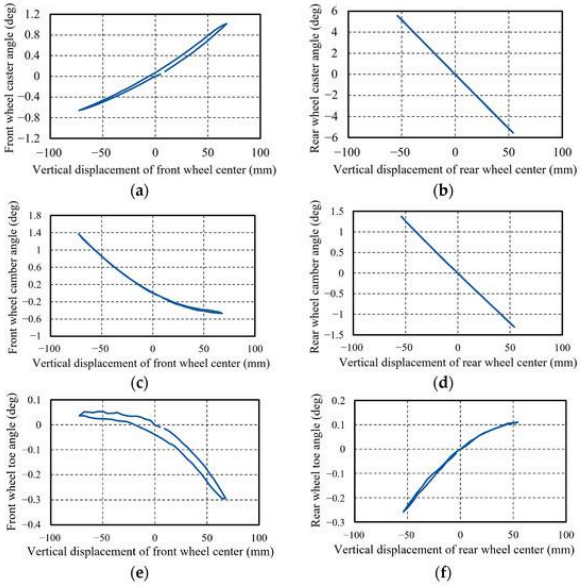
Gamma Technologies LLC



3D MBD suspension model

high fidelity 3D MBD model to generate suspension K&C curves

K&C curves used as inputs to represent suspension characteristics in simplified models

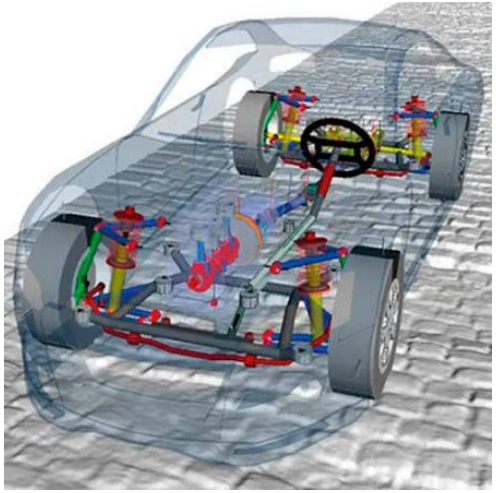


Suspension Kinematics and Compliance curves

Developed simulation tool that uses K&C parametric curves as input. The tool was integrated in a System level simulation suite (GT-ISE).

e) Vehicle Steering System Controls Integration

MSC Software/He



3D MBD vehicle model

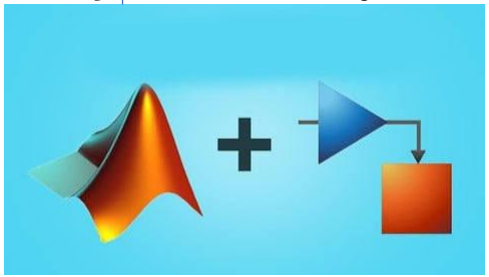
Vehicle States

Steering torque/Angle

FUNCTIONAL MOCK-UP INTERFACE

Steering torque/Angle

Vehicle States



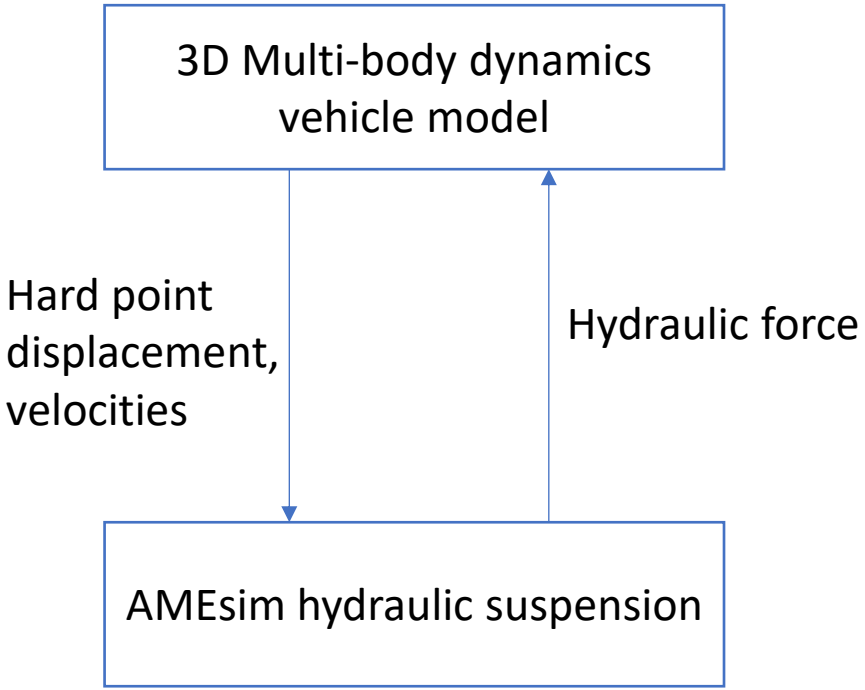
Simulink Controller

High fidelity 3D Vehicle Adams Model co-simulation with a Simulink steering system controller. Tools used: MSC Adams, Simulink, FMU co-simulation.

f) Sprayer Dynamics and Hydraulics Simulation

CNH Industrial





Test Data Correlated 3D Vehicle Model + hydraulic suspension model. 95% correlation.
Tools Used: Simcenter Motion, AMEsim

g) Skid Steer Inverse Dynamics

CNH Industrial



Inverse dynamics correlation of a 3D multi-body dynamics skid steer model
Tools Used: Simcenter Motion

Vehicle, Tilt Cylinder, and Lift cylinder displacements

↓

Track, actuator cylinder forces

SUMMARY

Why Work With Me

01

Leadership & Ownership

I like to take ownership of projects and lead them to success.



02

Humble & Amicable

People like collaborating with me because I am humble, and amicable.



03

Confident & Curious

I am confident enough to acknowledge gaps and explore new solutions that help overcome those gaps.



Let's Connect

Simulation Systems Architect — Available for Engagements & Collaborations


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srinivas-lakshmanan/ProjectPortfolio


Srinivas Lakshmanan | LinkedIn

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